4	**(a)(i)**	The volume decreases, the time between collisions with wall of cylinder is reduced / The frequency of collision with the walls of the cylinder increases. The average force on the wall / rate of change of momentum of molecules is increased. Hence, pressure increases.	**B1** **B1**
	(a)(ii)	1. $V \propto T$ Thus, $V_2 = V_1(T_2/T_1)$ $= (5.0 \times 10^{-3})\ (80 + 273.15)/(35 + 273.15)$ $= 5.7 \times 10^{-3}\ m^3$	**C1** **A1**
		2. $W = p\,\Delta V$ $= (1.0 \times 10^5)(5.7 \times 10^{-3} - 5.0 \times 10^{-3})$ $= 70\ J$	**A1**
	(b)(i)	$P = 230\ (8) = 1840\ W$	**A1**
	(b)(ii)	$Q = m\ l_v$ $1840\ (10) = (8.1 \times 10^{-3})\ l_v$ $l_v = 2.27 \times 10^6\ J\ kg^{-1}$	**C1** **A1**
	(b)(iii)	Water vapour condensing at the inner lid of the kettle and dripping back into the water. The decrease in the reading of the balance is less which causes the calculated value of l_v to be greater / an overestimation.	**B1** **B1**